

Timing the tide from space: intertidal topography from Sentinel-2 with the estuarine tide measured from the imagery itself

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Abstract

Satellite archives allow the topography of tidal flats, among the most extensive and most threatened coastal ecosystems, to be mapped by reading for every pixel the water level at which it floods. Every published implementation assigns each scene one water level from a tide model or gauge at one point. Inside an estuary that level is wrong: the tide arrives later and deformed the farther it travels up the channel, so the error grows inland and is systematic. MAREA measures this interior lag from the imagery itself. Each intertidal pixel is a binary tide gauge; for each 3-km band of along-channel distance the lag that best explains its ten-year wet/dry record maximises a profiled Bernoulli likelihood anchored at the mouth, the lags monotone by construction, each with a bootstrap interval. A binary archive measures the phase of the interior tide but not its amplitude or datum; only the phase is corrected. On four European estuaries with gauge pairs, MAREA measured lags of 63 and 129 min at the inner gauges of the Ems-Dollard and the Wadden Sea, where the gauges show 51 and 88, applied 7 min in the Westerschelde and none in the Ría de Ferrol, and cut the water-level error at the inner gauge by 42 and 46 % in the two large estuaries. Against the official Dutch bathymetry its maps reach 0.33 to 0.44 m RMSE with slopes of 0.65 to 0.81, against 0.48 to 0.78 m for the published logistic alternative; against an RTK-GNSS survey in the Ría de Villaviciosa (Spain) the estimator reaches 0.19 m RMSE with a slope of 0.72, against 0.27 m and 0.39 for the alternative. The method needs no instrument inside the estuary; code, notebooks and negative results are released.

Keywords: intertidal topography, tidal flats, Sentinel-2, tidal propagation, estuaries, waterline method, tide models, profile likelihood, digital elevation model

1. Introduction

Tidal flats (Fig. 1) are among the most extensive and least mapped coastal ecosystems: the first global inventory found at least 127 921 km² and a 16 % loss between 1984 and 2016 [1]. They are productive and buffer the coast [2], and their elevation governs flooding, habitat and surge exposure, yet coastal topography is coarse, outdated or absent over much of the world [3] because surveying a mudflat is slow, expensive and often dangerous [4, 2]. Global surface-water products ignore the tidal stage of each image [5, 6], the orbit samples the tide unevenly [7] and clear low-tide scenes are scarce: four usable Sentinel-2 images a year at Santander Bay, on the coast studied here [8].

The archive offers a way round the survey: the tide scans the relief, and a dated image series is a series of yes/no questions about every pixel's elevation. Waterline methods [9, 10] tag each scene's land–water contour with the water level at acquisition and interpolate between contours, with sub-pixel refinements [11, 12], and have produced continental models [2, 13, 6]. Per-pixel methods fit a logistic curve to each pixel's response against water level, its midpoint the elevation, from X-band SAR [14], Sentinel-2 near-infrared reflectance [15, 16] or the water/land class [17], or cross-correlate radar series with



Figure 1: Tidal flats at low water (German Wadden Sea, Wangerooge, August 2019). Left, a guided walk across flats that the next high water will cover; right, the edge of the flat at a groyne, with gulls and waders on the exposed mud. Photographs by Rhetos, Wikimedia Commons, CC0 1.0.

the wet/dry sequence of each candidate elevation [4]. These are the quantal-response ogives of Gaddum [18] and Bliss [19], the probit and logit of Finney and Berkson [20, 21]: a shore-line pixel is a mixture whose index tracks its water fraction [22, 23, 11], the cumulative distribution of its internal relief, with the elevation at its 50 % point.

All share one input: the water level of each scene, from a tide model or gauge at one point, applied to the whole scene or cell [2, 13, 15, 6, 16, 17]. The caveat is on record: a single instantaneous level “is no doubt a simplification of reality” [2]; accuracy falls with distance from the modelling point,

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with lags in estuaries [13]; part of a -0.51 m bias on the Tagus comes from “different tide height values for the same image acquisition time” [15]; phase differences exceed one hour within one Sentinel-2 scene [16]; a mean absolute error fell from 0.64 to 0.46 m with a closer gauge [17]; and a shore-based radar measured a bias of 0.12 m within 0.75 km of the gauge and about 0.5 m beyond [4]. Every reported survey accuracy, from ± 0.39 m [13] to 0.57 m [2], carries this error.

Global tide models cannot remove it (Fig. 2): they stop at the coast. FES2014 [24], EOT20 [25] and TPXO [26] reach about 2 cm in the open ocean but diverge near the coast [25]. EOT20 discards altimetry within 3 to 7 km of the coast and holds no observation inside an estuary; FES2014 resolves at best 1.5 km in estuaries, sees no water shallower than 10 m, down-weights coastal gauges and calls estuaries and deltas an open problem [24]. Inter-model differences exceed 30 cm near coasts, with metre-level errors at individual gauges [27], and the best regional ensemble still leaves 0.29 to 0.36 m at sheltered inlets, with “visible sharp elevation discontinuities” traced to a phase error [28]. Inside an estuary the tide arrives later the farther it travels, amplified by convergence, damped by friction and distorted in shallow water [29, 30], so a level read at the mouth is wrong inland by an amount that grows with distance and repeats in every scene.

Two studies read tidal information from the imagery itself. Granadeiro et al. [16] fit separate logistic curves to rising and ebbing scenes and take the lag that reconciles the two elevations, giving cotidal lines over the Bijagós archipelago within 6.6 min of the official values at five stations; it returns one lag per location, needs a reference site with a known tide and was tested on an open archipelago, not an estuary. Bishop-Taylor et al. [28] rank ten global tide models by the correlation of each pixel’s NDWI with the modelled tide, which reads phase but could rank a model highly “even if it greatly misrepresented tidal amplitude”, and leave estuarine amplification and damping uncovered. Neither estimates the interior lag with a statistical model of the pixel, states what a wet/dry archive can identify, or tests a measured lag against sampling noise.

This paper does those three things. MAREA treats every intertidal pixel as a binary tide gauge, a Rasch-type model with scenes as persons and pixels as items [31, 32], and measures the phase lag of the estuarine tide from the Sentinel-2 archive alone: per 3-km band of along-channel distance, the lag that best explains the ten-year wet/dry record maximises a profiled Bernoulli likelihood, each pixel’s elevation and width refitted under every candidate lag [33, 34], anchored at the mouth to a boundary that never includes the judge gauge. Binary data identify phase but not gain or datum; the lags are monotone inland by construction and each carries a bootstrap interval. Validation uses four European estuaries with gauge pairs, the official Rijkswaterstaat bathymetry at the Dutch sites, and the Ría de Villaviciosa (Spain) with an RTK-GNSS survey, against the method of Granadeiro et al. [16] reimplemented on the same pixels.

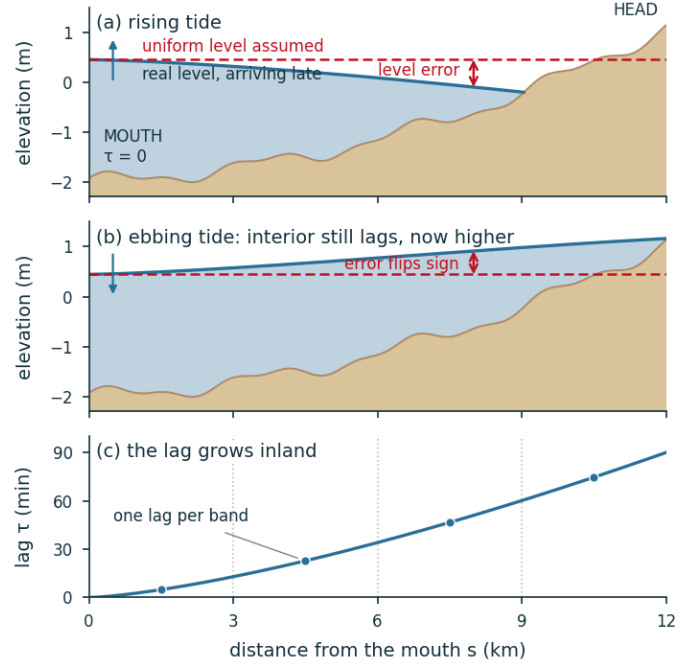


Figure 2: The error the method measures. (a) On a rising tide the water surface inside an estuary slopes down inland because the tide arrives late; the single level every published method reads at the mouth (dashed) is wrong by an amount that grows with distance. (b) On the ebb the interior still lags, so the sign of the error flips with the limb. (c) The lag τ grows with the distance s from the mouth; MAREA measures one lag per 3-km band of s .

2. Methods

2.1. Study sites and data

2.1.1. Sites

The method was developed on the Ría de Villaviciosa and its interior clock tested on four other estuaries (Fig. 3, Table 1). Villaviciosa is a 7 km drowned valley on the Cantabrian coast of Spain, one channel with mudflats on both banks; its ten-year archive at 10 m (1 379 scenes) was the test bed of every stage and holds the only field survey, an RTK-GNSS transect. The ría has no tide gauge, so its boundary is the ocean model alone, and the measured lag on the transect is a few minutes: Villaviciosa validates the estimator more than the clock.

The four test sites were chosen by one criterion: two tide gauges on one tidal axis, one at the mouth to feed the boundary and one inside to judge, never entering any fit. The Ría de Ferrol (Galicia), short and deep, is the negative control: the tide should arrive almost unchanged. The Westerschelde is the funnel mouth of the Scheldt between Vlissingen and Terneuzen; the Ems-Dollard (Netherlands and Germany) is a 35 km estuary whose inner basin, the Dollard, is a mudflat bay behind a constricted channel; the Wadden site is the Vlie basin between Terschelling and Harlingen.

Beyond the five sites, the pipeline ran unattended on 130 cells of 10 km along the north coast of Spain, from the Ría de Ribadeo to the Bidasoa (Fig. 4), 286 km² of intertidal ground with a median of 461 scenes per cell; under the detection rule then in force, 108 cells applied an interior lag and 21 kept the

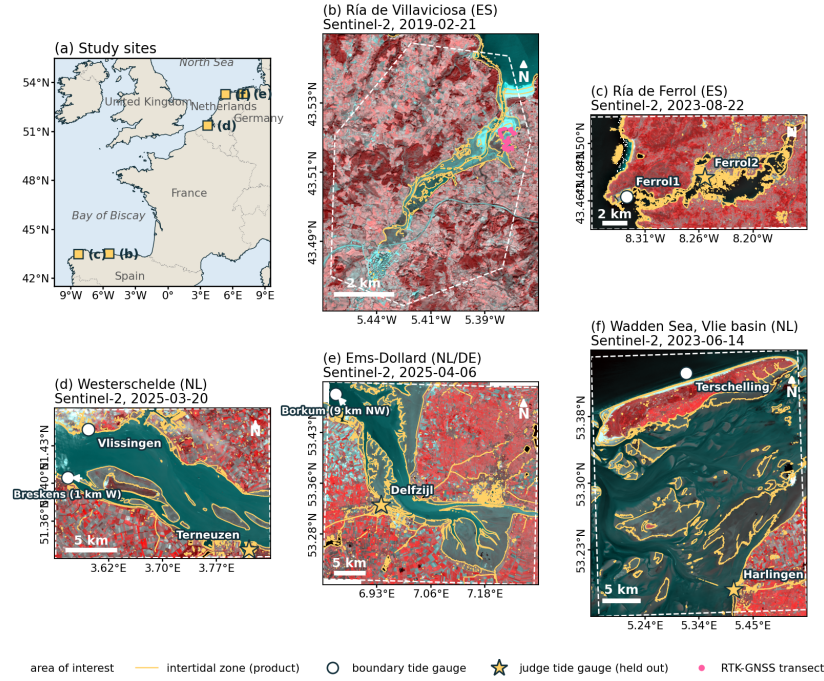


Figure 3: Study sites. (a) The five sites on the Atlantic and North Sea coasts of Europe (Natural Earth coastline). (b)–(f) Each site on a clear low-water Sentinel-2 scene from its datacube, as a false-colour composite (near-infrared, green, green: water dark, exposed sediment grey, vegetation red). Dashed line, area of interest; yellow outline, intertidal zone retained by the product; circles, tide gauges feeding the boundary; star, gauge held out as judge; a gauge outside the frame is marked at the nearest edge with its distance. In (b) the magenta points are the development subset of the RTK-GNSS survey. Scene dates are in the panel titles.

Table 1: Study sites. Scenes are all Sentinel-2 L2A acquisitions in the period at the stated resolution, before cloud screening; intertidal pixels have water frequency between 0.10 and 0.90 and at least 30 clear observations; the judge’s distance is the geodesic distance through the water from the sea edge of the site.

site	res.	period	scenes	intertidal px	boundary gauges	judge (km)
Ría de Villaviciosa (ES) ^a	10 m	2016–25	1 379	—	none (EOT20 only)	none
Ría de Ferrol (ES)	10 m	2023–25	461	103 759	Ferrol1	Ferrol2 (10.1)
Westerschelde (NL) ^b	20 m	2023–25	464	73 764	Vlissingen, Breskens	Terneuzen (21.9)
Ems-Dollard (NL/DE) ^b	20 m	2023–25	462	477 745	Borkum	Delfzijl (24.4)
Wadden, Vlie basin (NL) ^b	20 m	2023–25	462	367 076	Terschelling	Harlingen (18.3)

^a elevation truth: RTK-GNSS survey. ^b elevation truth: Rijkswaterstaat Vaklodingen.

uniform level, with a median largest lag of 24 min per cell. The census is released with the code; its cells are not validated here.

2.1.2. Sentinel-2 archive

All imagery is Sentinel-2 Level-2A surface reflectance [35], processed with Sen2Cor [36], from the Copernicus Data Space Ecosystem via openEO. Every acquisition of the period, with no cloud selection, enters a datacube of three layers per scene, the bands B03 (green) and B08 (near-infrared) at their native 10 m and the scene classification layer SCL; Villaviciosa and Ferrol stay at 10 m, the much larger Dutch sites are resampled to 20 m at download. The water index is the NDWI of McFeeters [37], $(B03 - B08) / (B03 + B08)$, preferred to its short-wave infrared variant [38] because both bands are native 10 m and SWIR indices read wet sediment as water [11, 28]. Acquisition instants come from the catalogue metadata, not the

nominal overpass time; the difference reaches minutes, hence centimetres of tide.

One openEO batch job per site builds the cube once, server-side; every later stage of the released library streams it from disk in time blocks of bounded memory, so a ten-year archive fits a laptop and cells run in parallel. Scenes are screened by the cloudy share of the transition zone, the pixels neither always land nor always water, not of the whole frame, and within a kept scene every cloudy pixel is masked by the scene classification (Fig. 5): a cloud over the hills costs nothing, a cloud over the flat costs observations, not the scene. At Villaviciosa 305 of 1 379 scenes pass the 10 % rule over the transition zone; the same rule over the frame would keep 307, but a different set: 25 scenes with a cloudy frame and a clear flat enter, 27 with a clear frame and a cloudy flat leave. On the Dutch frames, mostly sea and flat, the two rules nearly coincide (66 against 67

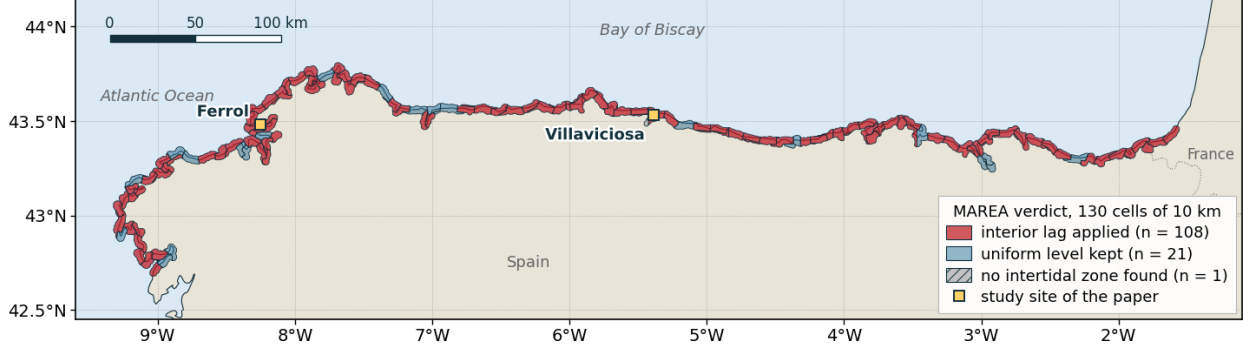


Figure 4: The north-coast campaign: 130 cells of 10 km along the Cantabrian and Galician coast of Spain, coloured by the verdict of the unattended run (interior lag applied, or uniform level kept), with the two Spanish study sites marked.

scenes on the Westerschelde, 48 against 45 on the Ems); there the frame rule left the comparison method 11 rising scenes on the Westerschelde, one short of the 12 its fit needs.

2.1.3. Tides

The ocean boundary is EOT20 [25], evaluated with pyTMD [39, 40] at a fixed point in each mouth channel at each acquisition instant. Tide-gauge records, minute-scale for 2023–2025 from the IOC Sea Level Station Monitoring Facility [41], are cleaned of spikes and outliers by a rolling-median rule, demeaned, and left with their gaps over one hour open; the Vlissingen record is missing for eighteen months. A boundary is the model alone, the N nearest gauges with inverse-square-distance weights, or their consensus, the plain mean of two; the choice is stated with every result, and a site’s judge gauge never enters its boundary.

2.1.4. Elevation truth

For the Dutch sites the truth is the Vaklodingen of Rijkswaterstaat [42], grids from echo-sounding and airborne laser altimetry at 20 m relative to NAP; 32 sheets of different survey years cover the three sites, the Dollard last in 2020. For Villaviciosa it is an RTK-GNSS survey: 361 fixed solutions of 1.3 cm nominal precision along a 0.75 km transect at the lowest spring tide of the month. Before any elevation product existed, 35 % of the surveyed pixel blocks were sealed under a runtime guard; all numbers here use the development subset. The 5 m national LiDAR (PNOA) is a second reference at the two Spanish sites, with the caveat of Section 2.6: over the flats it records the water surface at flight time. All truths are in datums other than the tide model’s, so every comparison removes and reports the median offset.

2.2. The pixel model and the elevation inversion

For a shoreline pixel with flooded fraction f , linear mixing of the McFeeters water index [37] gives

$$\text{NDWI} = (1 - f)\text{NDWI}_{\text{dry}} + f\text{NDWI}_{\text{wet}} = a + b f, \quad (1)$$

where the dry level a and the dry-to-wet jump b are fitted per pixel, which normalises albedo between pixels. At water level

h , $f(h) = \text{Pr}(\text{internal elevation} \leq h)$ is the cumulative distribution of the pixel’s relief, assumed Gaussian with mean z , its median elevation, and spread σ , its roughness:

$$\text{NDWI}(h) = a + b \Phi\left(\frac{h - z}{\sigma}\right), \quad (2)$$

with $\Phi(u) = (2\pi)^{-1/2} \int_{-\infty}^u e^{-v^2/2} dv$ the standard normal distribution function; the midpoint $h = z$ is the elevation. Eq. (2) is the probit ogive of quantal-response analysis [19, 20], the Gaussian counterpart of the per-pixel methods’ logistic [21]. The fitted σ also absorbs the water-level error of each scene.

The inversion (`invert_series`) fits Eq. (2) to each pixel’s clear observations by least squares: for each candidate (z, σ) the linear (a, b) is solved in closed form, z runs over 50 points between the lowest and highest level of the series, σ over the archive’s eight relief atoms (0.03, 0.06, 0.10, 0.15, 0.22, 0.32, 0.45 and 0.65 m), and the pair with the smallest residual is kept. A pixel is left blank with fewer than 8 clear observations, a contrast b outside $(0.15, 2.5)$ or a dry level $|a| \geq 2.5$.

A scene is kept if at most 10 % of the transition zone is cloudy; an observation is used only where Sen2Cor reads vegetation, bare ground or water (the campaign extraction also admits unclassified). The wet/dry label is $\text{NDWI} > 0$: Otsu’s threshold [43] on the twelve clearest development-site scenes lands at +0.25, the top of its allowed range; zero is preferred because turbid estuarine water reads lower than clear water. The intertidal zone is delimited by water frequency, the wet fraction of a pixel’s clear observations. A three-class Otsu split of the transition pixels separates ground, flat and channel, with valleys at 0.21 and 0.66 in Villaviciosa, and confirms that the archive holds three populations; but the flats reach far into the rarely wet class: of the 181 intertidal pixels the development RTK survey covers, only 79 lie inside the Otsu window against 130 inside 0.05–0.95. The window is therefore set by what the inversion can resolve, a pixel with enough clear observations on both sides of its transition, and opened to 0.05–0.95. The four test-site products use the fixed window and minimum count of Section 2.1. Elevations are fitted within an epoch, 2023–2025 for the products of Section 3.

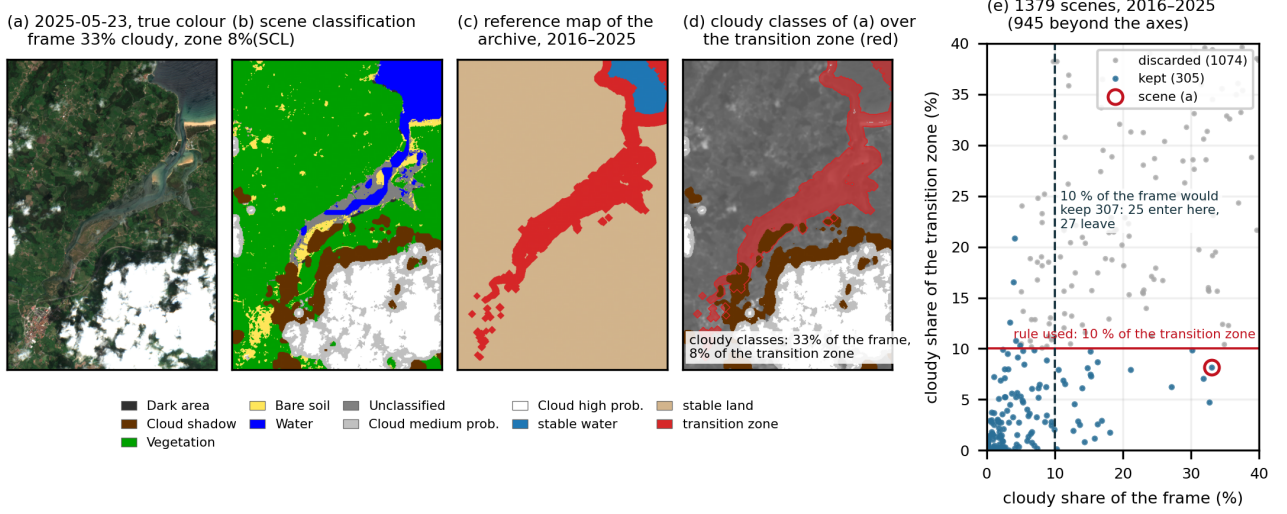


Figure 5: Cloud screening over the transition zone, Ría de Villaviciosa. (a) A kept scene in true colour, clouds over the hills and a clear ría; (b) its Sentinel-2 scene classification; (c) the reference map of the archive: stable water, stable land and the transition zone between them, the only pixels the screening looks at; (d) the cloud mask of that scene laid over the transition zone: the share of the zone under cloud is what the rule reads; (e) every scene of the archive by the cloudy share of the frame and of the transition zone, with the 10 % rules and the number of scenes each keeps.

2.3. The interior clock

The interior tide is measured per band of along-channel distance s from the mouth: s is the geodesic distance, the minimum-cost eight-connected path over the permanent-water and intertidal masks times the pixel size, from the permanent-water pixels (water frequency at least 0.90) on the image border on the sides declared as sea. Intertidal pixels are split into 3-km bands along s ; a band with fewer than 500 pixels is merged with the next. Each band sits at the median s of its pixels, and at most 1 200 of them, drawn at random with a fixed seed, enter the likelihood.

Each observation is reduced to its binary label $y_{pt} \in \{0, 1\}$ and the amplitudes of Eq. (2) are fixed at 0 and 1, leaving two parameters per pixel: a Rasch-type model [31, 32] (scenes as persons, pixels as items). Under a candidate clock τ_k for band k the level at scene t is $h(t - \tau_k)$, pixel p is wet with probability $P_{pt} = \Phi((h(t - \tau_k) - z_p)/\sigma_p)$, and the profiled Bernoulli log-likelihood of the band is

$$\ell_k(\tau_k) = \sum_{p \in k} \max_{z_p, \sigma_p} \sum_t [y_{pt} \log P_{pt} + (1 - y_{pt}) \log(1 - P_{pt})], \quad (3)$$

where t runs over the pixel's clear scenes. The inner maximum is the profiling of variable projection [33, 34]: every pixel re-fits its own (z_p, σ_p) under each clock, so clocks are decided only by what terrain cannot absorb, the reordering of the tide across dates (Fig. 8). The grid is 60 points of z_p over the boundary's tidal range plus 0.3 m on either side, about 8 cm apart, and, in the product, the six interior relief atoms, 0.06 to 0.45 m, for σ_p ; probabilities are clipped at 10^{-4} , and a pixel needs at least six clear scenes.

The boundary is evaluated once at every acquisition instant shifted by the sixteen values $-45, -30, \dots, 180$ min of the shift bank, and $h(t - \tau)$ is interpolated linearly between them. The mouth band is pinned at $\tau_0 = 0$. The free parameters are the

increments between successive bands, $\tau_k = \sum_{j \leq k} \delta_j$ with every $\delta_j \geq 0$, so the profile is monotone non-decreasing inland. The negative of Eq. (3), summed over bands and normalised per observation, is minimised with L-BFGS-B [44] in SciPy [45], the increments in hours, bounded between 0 and +180 min and started at zero.

2.4. Identifiability and adoption

The interior tide may also be amplified or damped: with a gain α and a datum shift, $h_{\text{int}}(t) = \alpha h(t - \tau) + \text{datum}$. Binary data cannot read α : with $P = \Phi((\alpha h - z)/\sigma)$, multiplying α , z and σ by one constant c gives

$$\frac{(c\alpha)h - (cz)}{c\sigma} = \frac{\alpha h - z}{\sigma}, \quad (4)$$

so the likelihood is unchanged, and a datum shift is absorbed by z . Only the phase survives, so it is measured from the imagery; gain and datum come from the boundary. The check is executable: with a gain of 1.0–1.3 planted in the simulator, the negative log-likelihood profile in α , with (z, σ) profiled on grids scaling with α , must be flat.

Each band lag carries a scene-bootstrap interval: the scenes are resampled with replacement thirty times, the profile refitted on each, and the 2.5th and 97.5th percentiles of the thirty τ_k are its 95 % confidence interval; an interval containing zero means the lag is not distinguishable from none. No threshold is applied and no lag discarded; the interval says which values carry information. On replica archives with the pixels, cloud masks and acquisition instants of a real site but a uniform tide with no interior lag, the estimator returns lags of the order of ten minutes; the experiments are released with the code.

Each pixel receives the band profile smoothed along s with a Gaussian kernel of the band length, anchored at $\tau = 0$ at the mouth and kept monotone, evaluated at the pixel's own distance

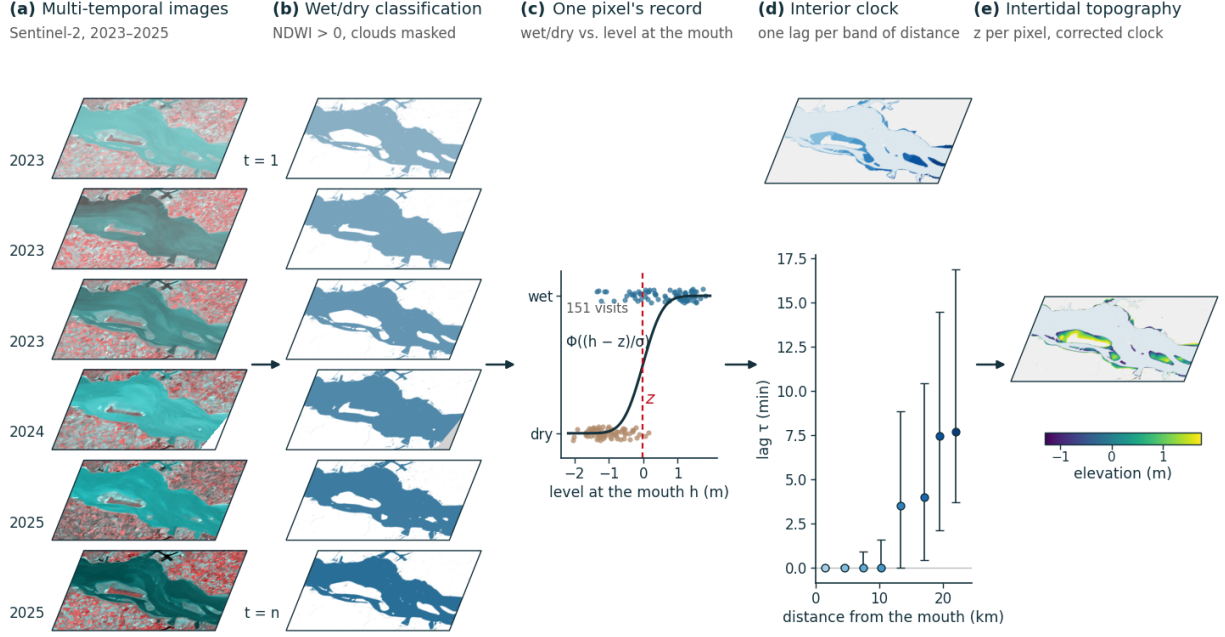


Figure 6: The pipeline on the Westerschelde, 2023–2025. (a) Dated scenes of the archive; (b) each scene reduced to wet/dry, NDWI above the threshold on clear-sky pixels; (c) one pixel’s record against the water level at the mouth at each visit, with its fitted transition; (d) the interior clock: bands of geodesic distance from the mouth and one lag per band with its bootstrap interval; (e) the elevation map, inverted pixel by pixel on the corrected clock.

and rounded to the minute. The inversion of Section 2.2 is then run once with the boundary evaluated at the exactly shifted instants $t - \tau$, not the bank’s 15-min interpolation, and the z grid spanning the unshifted boundary range everywhere. The product stores the band lags, their intervals and the lag every pixel received.

2.5. The comparison method

The elevation maps are compared with those of the published method of Granadeiro et al. [16], which fits a logistic with four parameters to standardised near-infrared reflectance against water height per pixel, the inflection being the elevation, and reads a tidal-stage lag per pixel from the disagreement between its rising and ebbing fits, extended over the flat by a thin-plate spline. It was reimplemented from the paper, not their code, with SciPy [45] for the fits, and run on the same pixels, scenes and boundary as MAREA; the reimplementation and its declared substitutions are released with the code.

2.6. Validation

The method yields two quantities, the lag of each band and the elevation of each pixel, each validated on site, in the Ría de Villaviciosa, and externally, against tide-gauge records and surveys made by others (Table 2). The lag blocks ask whether the measured lag is real and whether applying it improves the level assigned inside the estuary; the elevation blocks set both maps against independent surveys. Nothing that judges a result enters a fit.

2.6.1. Metrics and the datum

Lags are compared in minutes: the gauge-pair lag is the mean over paired tidal cycles c of $t_c^{\text{in}} - t_c^{\text{mouth}}$, the instants at which the inner and the mouth record cross the same level on the same limb, and it is set against the lag MAREA applies at the inner gauge. The effect of a lag on the water level is the RMSE between the level a product predicts there, the boundary h_B read τ minutes earlier, and the level h^{in} the gauge recorded, the median of the difference (the datum) removed:

$$\text{RMSE}_h = \sqrt{\frac{1}{N} \sum_t (d(t) - \text{median } d)^2},$$

$$d(t) = h_B(t - \tau) - h^{\text{in}}(t), \quad (5)$$

over the N ten-minute samples of 2023–2025. Elevations are compared with Eq. (6). Products are on the datum of the tide model at the mouth, the mean sea level of EOT20 [25]; each truth is on its own: NAP for the Vaklodigen [42], the ellipsoid for the RTK-GNSS survey, orthometric heights for the LiDAR. No method can see the constant between the two, so every comparison removes it and reports it as the bias. On the n samples where product z and truth z^t are both finite, each series is centred on its own median, $y_i = z_i - \text{median}(z)$ and $x_i = z_i^t - \text{median}(z^t)$, and

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2}, \quad b = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2},$$

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}, \quad (6)$$

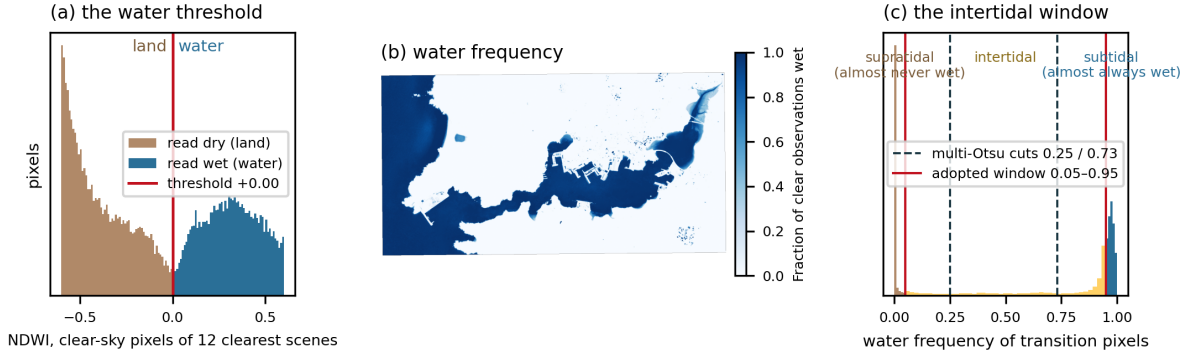


Figure 7: Water detection on the Ría de Ferrol, 2023–2025. (a) NDWI of the clear-sky pixels of the twelve clearest scenes and the adopted threshold: land dominates the histogram and turbid estuarine water sits below the textbook 0.1, so the split is set at 0 at every site; (b) the water frequency, framed by the study polygon; (c) the water frequency of the transition pixels, the multi-Otsu cuts between the three classes and the adopted window, wider than the cuts because the surveyed flats of Villaviciosa lie mostly in the rarely wet class (Section 2.2).

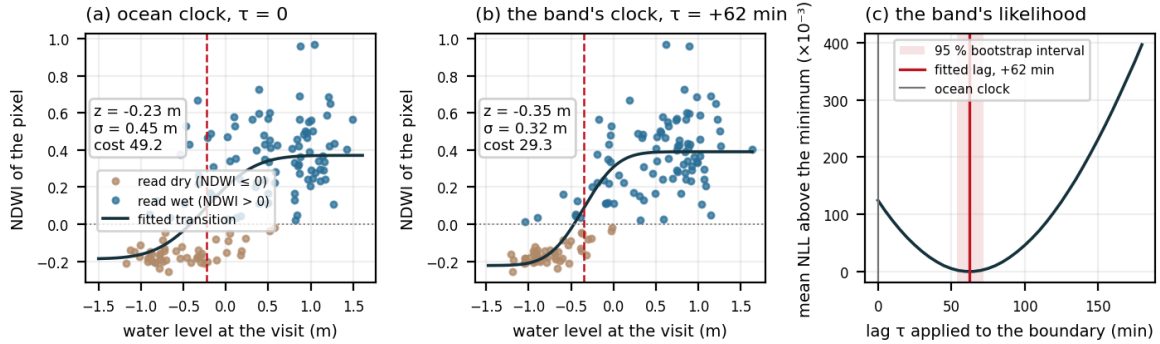


Figure 8: What the clock does to the likelihood, on the Ems-Dollard band containing the Delfzijl gauge. (a) One pixel of the band on the ocean clock: its NDWI at each visit against the water level at the mouth, coloured by the reading (wet above 0, dry below), and the transition $a + b\Phi((h - z)/\sigma)$ with the (z, σ) profiled by the binary likelihood; wet and dry readings overlap over a wide range of levels. (b) The same pixel on the band's clock: the readings separate, the transition sharpens and the pixel's negative log-likelihood (cost) falls. (c) The band's profiled negative log-likelihood against the lag, relative to its minimum, with the fitted lag, its 95 % bootstrap interval and the ocean clock.

with the truth on the abscissa: the slope b says whether the map reproduces the relief (1) or compresses it (below 1), r how much of it the map explains. Only truth within the range of scene levels, widened by ± 0.25 m, is scored: no waterline method can represent a channel bed at -8 m beside a flat. A 5×5 thinning of the grid, one pixel per 100 m block, changed no ranking.

2.6.2. The interior lag on site: pressure sensors along the ría

The Ría de Villaviciosa has no interior tide gauge, so its lag will be measured directly: absolute pressure sensors at three to four stations, submerged through the lowest spring low water, with a barometer on land, over 29 days at 1 to 5 min. Level follows $h = z_s + (p - p_{\text{atm}})/(\rho g)$, with $\rho = 1025 \text{ kg m}^{-3}$ and z_s the sensor elevation from RTK-GNSS. Each station is then scored against the mouth as the gauge pairs below. The campaign is pending.

2.6.3. The interior lag against gauge pairs

Four estuaries have a gauge near the mouth and another inside, the judge of Section 2.1: Ferrol1 and Ferrol2, Vlissingen and Terneuzen, Borkum and Delfzijl, Terschelling Noordzee

and Harlingen. Two quantities need only the two records [41], 2023–2025, 10-min grid, demeaned, gaps over one hour left open. The first is the lag in reaching the same level: crossings of the 25th, 50th and 75th percentile of the mouth's level are interpolated between samples and paired with the nearest inner crossing within three hours; their mean difference over paired cycles, and per limb, is the gauge-pair lag. It is compared with the lag MAREA applies at the gauge's distance and with the 95 % bootstrap interval of the band containing the inner gauge, measured from band 0, which holds the mouth gauge (*no delay*: 0 everywhere). The second is the same-instant level difference, inner minus mouth: its RMS and mean are the error of the uniform level every published method assumes. The third is the level predicted at the inner gauge from the mouth boundary read τ minutes earlier, with $\tau = 0$ and with MAREA's lag there, scored on the gauge's record as the median-centred RMSE of Eq. (6). Nothing was fitted on the inner gauge, so the whole record is the test. The comparison method's lag, per pixel over the flats, reaches no gauge and is scored only through its elevations.

Table 2: The four validation blocks: what is validated, against what, where, and with which metric.

block	reference	sites	metric
lag, on site	pressure sensors along the ría (campaign pending)	Villaviciosa	lag in reaching the same level (min); same-instant level difference (m)
lag, external	gauge pair: mouth and inner (judge)	Ferrol, Westerschelde, Ems-Dollard, Wadden	gauge-pair lag vs. lag applied (min); RMSE of the level at the judge, without and with the lag (m)
elevation, on site	RTK-GNSS survey, development split	Villaviciosa	RMSE, slope b , r of each map on the survey (Eq. 6); transect profile
elevation, external	IGN LiDAR DTM; Vaklodingen	Spanish sites; Dutch sites	RMSE, slope b , r (Eq. 6), by band for the Vaklodingen

2.6.4. Elevation on site: the RTK-GNSS survey

In Villaviciosa the maps are scored on the development subset of the RTK-GNSS survey of Section 2.1: 35 % of its 150 m blocks, drawn once with a fixed seed and sealed before any elevation product existed, released by a runtime guard only to the verdict script. The two maps share record, epoch (2023–2025) and boundary (EOT20 at the mouth): *MAREA*, our inversion with the smoothed band profile at each pixel’s distance, and *Granadeiro*, the method of Section 2.5 with its own lag map. Both are scored with Eq. (6) and drawn with the survey along a straight stretch of the walked route.

2.6.5. Elevation external: the national LiDAR and the Vaklodingen

At the Spanish sites, Villaviciosa and Ferrol, the reference is the 5 m LiDAR DTM of the Instituto Geográfico Nacional (PNOA), resampled bilinearly onto the product grid; at the Dutch sites, the Rijkswaterstaat Vaklodingen [42], yearly bathymetry and intertidal topography on a 20 m grid (NAP), each pixel taking its nearest cell’s latest survey of 2019–2025. Both maps are scored with Eq. (6) on the intertidal pixels each resolves and on those both resolve, the Vaklodingen also by band of distance from the mouth. A LiDAR flown at high water records the water surface over the flats, so each LiDAR score comes with the water share of its truth: the share of intertidal cells within 5 cm of the modal value.

3. Results and discussion

3.1. The interior lag against the gauge pairs

The gauge pairs order the sites as the archive does (Table 3, Fig. 10). In Ferrol the inner gauge lags the mouth by 4 min and one level for the whole ría errs by 0.10 m RMS; *MAREA* measures 0 min there, interval 0 to +2, and the level error stays at 0.112 m. In the Westerschelde the gauges are 19 min apart on both limbs; *MAREA* applies +7 min at Terneuzen, the band’s interval reaching +17, and the error falls from 0.356 to 0.311 m. In the Ems-Dollard the lag is 51 min, 41 rising and 61 falling; *MAREA* applies +63 min at Delfzijl, inside its interval of +54 to +72, and the error falls from 0.620 to 0.358 m, by 42 %. In the Vlie basin the gauges are 88 min apart, 91 rising and 85 falling; *MAREA* applies +129 min at Harlingen, interval

+123 to +138, 41 min beyond the gauges, and the error still falls from 0.562 to 0.303 m, by 46 %. Two columns of Table 3 differ in kind: the same-instant difference sets the two gauge records against each other, whereas the level error is that of the boundary the products read, the consensus of EOT20 and the mouth gauge, which adds the model’s error to the uniform-level assumption; hence the boundary without a lag (0.620 m in the Ems-Dollard) is worse than the mouth gauge copied inland (0.50 m), and *MAREA* with the lag (0.358 m) beats either. Where an interior tide exists the archive measures it; where none exists it applies the few minutes its intervals declare, at a cost of centimetres at most.

3.2. Elevation against the RTK survey

On the 194 surveyed points it resolves, *MAREA* reaches 0.193 m RMSE with slope 0.72 and $r = 0.86$; the *Granadeiro* map reaches 0.275 m with slope 0.39 and $r = 0.63$ on 163 points (Table 4, Fig. 11a,b). Along a straight 192-m stretch of the walked route (Fig. 11c–e) the *MAREA* profile follows the survey within the pixel size, including the fall at its inland end, where the *Granadeiro* profile stays flat. The lag applied on the surveyed pixels is 5 min in median and 16 at most, so Villaviciosa tests the estimator more than the clock.

3.3. Elevation against the external references

Against the Vaklodingen (Table 5, Fig. 12) *MAREA* is the closer map at the three Dutch sites. In the Westerschelde it scores 0.435 m RMSE with slope 0.76 and $r = 0.92$ on 62 000 pixels, *Granadeiro* 0.777 m on 45 000. In the Ems-Dollard *MAREA* scores 0.363 m with slope 0.65 and $r = 0.84$ on 413 000 pixels; *Granadeiro* resolves a quarter of them at 0.531 m with slope 0.39. In the Vlie basin *MAREA* scores 0.329 m with slope 0.81 on 349 000 pixels, *Granadeiro* 0.477 m on 66 000; on the pixels both resolve, 0.230 against 0.427 m. Fig. 9 shows the Vlie map beside the Vaklodingen and the hydroperiod derived from it. The lag carries the two large sites: inverted without it, the same pixels score 0.565 m with slope 0.24 in the Ems-Dollard and 0.474 m in the Vlie, and beyond 28 km of the Ems the error falls from 0.50–0.66 m to 0.28–0.42 m with the clock. At the Spanish sites the LiDAR is a water surface: 78 % of the Ferrol cells lie within 5 cm of 0.00 m, and inside the tide range both maps score 0.73 m there with correlations near

Table 3: The interior lag against the gauge pairs, 2023–2025. Lag between the mouth gauge and the inner gauge in reaching the same level (mean over paired cycles and limbs, positive: the inner gauge is later); lag MAREA applies at the inner gauge, with the 95 % bootstrap interval of its band; RMS difference between the two gauge records at the same instant, the error of copying the mouth gauge inland; and the RMSE of the level predicted at the inner gauge from the boundary by the products use (the consensus of EOT20 and the mouth gauge), without and with the lag.

site (inner gauge, km from mouth)	lag between gauges (min)	lag applied by MAREA (min)	inner vs. mouth gauge, same instant, RMS (m)	level at the inner gauge from the boundary (m): no lag → MAREA
Ferrol (Ferrol2, 10.1)	+4	0 [0, +2]	0.10	0.112 → 0.112
Westerschelde (Terneuzen, 21.9)	+19	+7 [+4, +17]	0.28	0.356 → 0.311
Ems-Dollard (Delfzijl, 24.4)	+51	+63 [+54, +72]	0.50	0.620 → 0.358
Wadden, Vlie (Harlingen, 18.3)	+88	+129 [+123, +138]	0.53	0.562 → 0.303

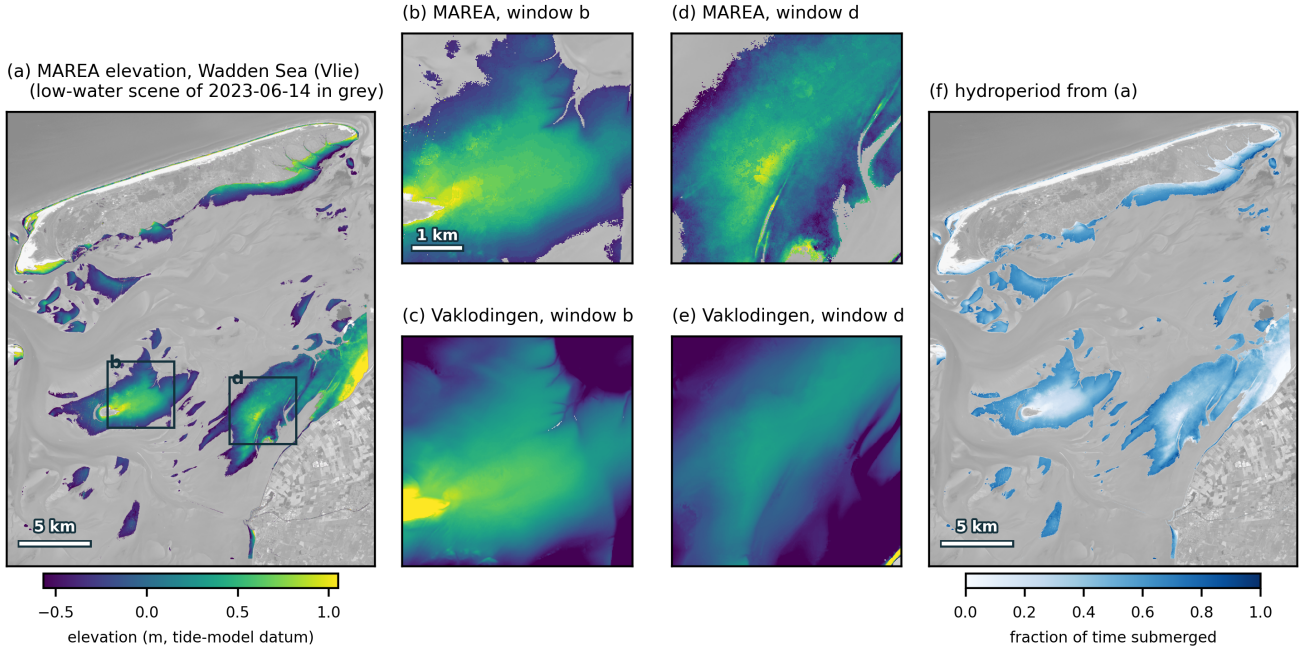


Figure 9: The products of the Wadden Sea site (Vlie inlet to Harlingen), 2023–2025, over the clearest low-water scene of the archive in grey. (a) The MAREA elevation map on the intertidal zone, with two zoom windows marked; (b), (d) the windows in the MAREA map (20-m grid); (c), (e) the same windows in the Rijkswaterstaat Vaklodigen, the official survey of the flats and channels, one value per 20-m cell taken cell for cell without interpolation, same colour scale, datum offset removed; (f) the hydroperiod, the fraction of time each pixel is submerged, from map (a) and the tide series.

Table 4: Elevation against the RTK-GNSS survey, Villaviciosa development subset, 2023–2025, EOT20 boundary. n : surveyed points the map resolves; RMSE, slope b and r as in Eq. (6), RTK on the abscissa; bias: the datum offset, ellipsoidal heights against the tide-model datum, removed before scoring.

method	n	RMSE (m)	b	r	bias (m)
MAREA	194	0.193	0.72	0.86	−52.8
Granadeiro	163	0.275	0.39	0.63	−53.3

0.1; in Villaviciosa 69 % of the 32 120 cells lie within 5 cm of +2.00 m, above the tide range and excluded, and on the 8 630 cells left both maps score 0.78 to 0.80 m with r below 0.35. The reference discriminates nothing at either site.

3.4. Limitations and future work

Gain.. A wet/dry archive fixes the phase of the interior tide, not its amplitude (Section 2.4): gain enters only as an external prior, and a strongly amplified tide leaves it unmeasured in the elevations.

Sampling.. A sun-synchronous sensor never sees S_2 and samples M_2 at a 14.8-day alias; the lowest spring tides never enter the archive, so the products stop at the lowest observed level.

Bands and smoothing.. The lag is measured per 3-km band and applied through a kernel of the same length, so sharper phase changes are not resolved. Lags are applied wherever fitted, noise included, at a cost of centimetres where no lag exists.

Boundary.. Lags are measured against the boundary series at the mouth, whose phase error is absorbed into the lag while the residual level error holds the surge and harmonics it lacks; the nearest gauge always beats the model and their consensus.

Epochs and truth years.. Morphology moves, so products are fitted per epoch of two to three years; the truths have their own dates (the Dollard, 2020).

The comparison method.. Granadeiro et al. [16] was reimplemented from the paper with declared substitutions; the numbers are ours, not theirs.

Table 5: Elevation against the external references: the IGN 5 m LiDAR DTM (PNOA) at the Spanish sites, the Rijkswaterstaat Vaklodigen at the Dutch sites. Each map on all intertidal pixels it resolves (left) and on those both resolve (right), Eq. (6), reference inside the tide range the maps could sample. Spanish site headers give the LiDAR diagnostic: modal value over the intertidal mask and share of cells within 5 cm of it, the water-surface share of the truth.

method	all resolved pixels				common pixels			<i>r</i>
	<i>n</i>	RMSE (m)	<i>b</i>	<i>r</i>	RMSE (m)	<i>b</i>		
<i>Ría de Villaviciosa</i> , IGN LiDAR — mode +2.00 m, 69 % of cells within 5 cm, above the tide range and excluded								
MAREA	7 458	0.803	0.47	0.33	0.760	0.42		—
Granadeiro	4 477	0.775	0.32	0.25	0.767	0.33		—
<i>Ría de Ferrol</i> , IGN LiDAR — mode 0.00 m, 78 % of cells within 5 cm								
MAREA	51 716	0.736	0.46	0.13	0.648	0.53		—
Granadeiro	49 262	0.732	0.36	0.09	0.644	0.54		—
<i>Westerschelde</i> , Vaklodingen								
MAREA	62 181	0.435	0.76	0.92	0.385	0.80		0.94
Granadeiro	45 005	0.777	0.78	0.77	0.776	0.78		0.78
<i>Ems-Dollard</i> , Vaklodingen								
MAREA	412 565	0.363	0.65	0.84	0.327	0.67		0.86
Granadeiro	98 953	0.531	0.39	0.58	0.521	0.40		0.58
<i>Wadden Sea (Vlie)</i> , Vaklodingen								
MAREA	348 894	0.329	0.81	0.68	0.230	1.00		0.83
Granadeiro	65 576	0.477	0.65	0.44	0.427	0.77		0.53

4. Conclusion

Every waterline product assigns a scene one water level; inside an estuary that level is wrong by the interior lag. A decade of Sentinel-2 wet/dry readings measures that lag with no instrument in the estuary: one profiled likelihood per band of distance from the mouth, monotone by construction, with a bootstrap interval. At four European estuaries the measured lags follow the held-out gauges, 0 min against 4, 7 against 19, 63 against 51 and 129 against 88, and applying them cut the water-level error at the inner gauge by 13, 42 and 46 % where a lag exists. Against the Dutch Vaklodigen the MAREA maps are closer at all three sites, 0.33 to 0.44 m RMSE with slopes of 0.65 to 0.81, against 0.48 to 0.78 m for the published logistic alternative; the RTK-GNSS survey of Villaviciosa gives 0.19 m against 0.27 m. What the archive cannot measure, the gain and datum of the interior tide, is stated as such. Code and notebooks are released; a pressure-sensor campaign will close the on-site validation.

Code and data availability

The library (pyintertidal), the executed pipeline notebooks, the validation scripts, the pre-registered configurations and every result file are released at [repository and archive DOI — TO FILL]. The code regenerates the inputs from open sources (Sentinel-2 L2A through openEO, IOC gauge records, the Vaklodigen, the PNOA LiDAR, EOT20); the development split of the RTK-GNSS survey is supplied, the reserved split stays sealed under a runtime guard.

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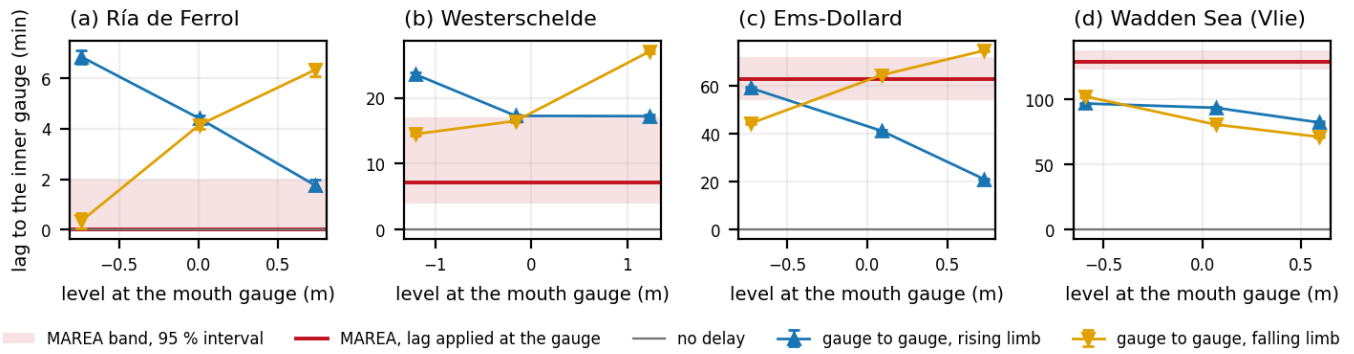


Figure 10: The interior lag against the gauge pairs. Lag between the mouth gauge and the inner gauge in reaching the same level, 2023–2025, at the 25th, 50th and 75th percentile of the mouth's level, rising and falling limb (mean over paired cycles, bars: standard error); the lag MAREA applies at the inner gauge (line) with the 95 % bootstrap interval of its band (shading); no delay at zero.

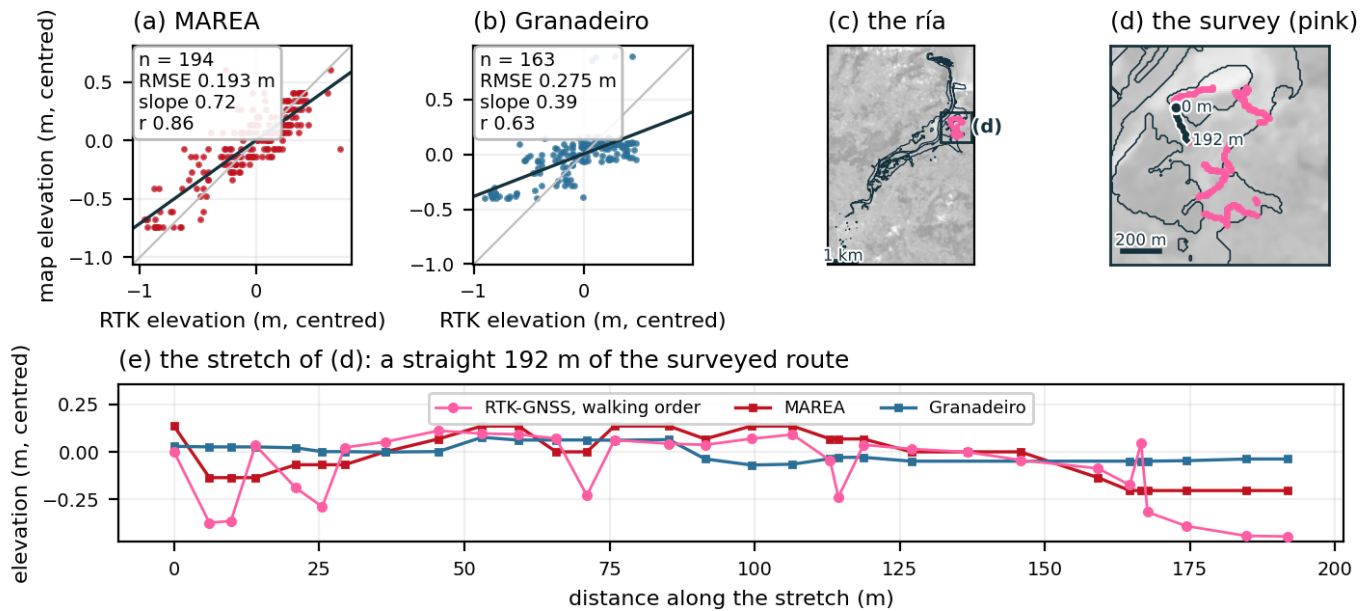


Figure 11: Elevation against the RTK-GNSS survey, Villaviciosa development subset. (a), (b) Each map on the RTK elevation, both centred on their medians, with the OLS line (dark) and the 1:1 line (grey). (c) The Ría on the low-water scene, intertidal zone outlined, surveyed points in pink; (d) the survey, with the straight stretch of (e) in white and its start at 0 m. (e) Profile along that stretch: the RTK points joined in walking order and the two maps at the pixels the route crosses, each series centred.

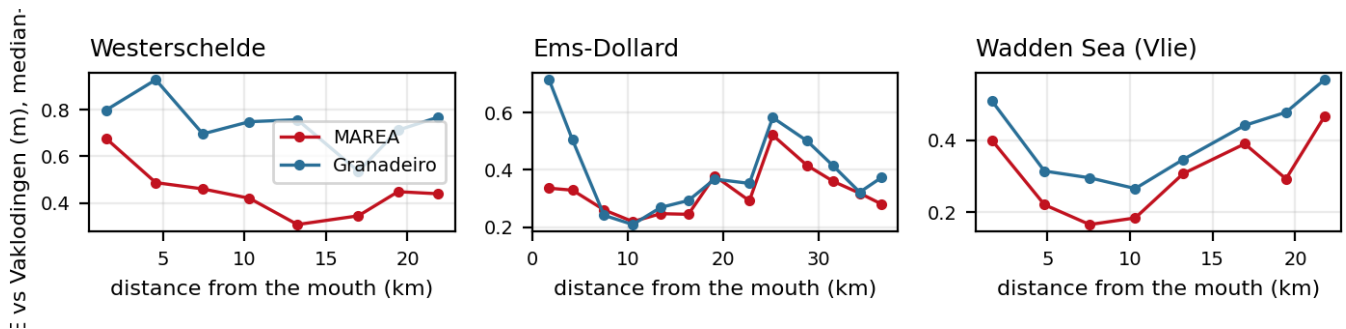


Figure 12: RMSE against the Vaklodigen by band of distance from the mouth, median-centred, for the two maps at the three Dutch sites.